

A Framework for Generalizable Classification Under Label Shift

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Background

- Classification learns a rule to predict labels from features.
- Standard methods assume source and target data come from the same distribution.
- In reality, future data often change across time or populations. This is called **dataset shift**.
- Dataset shift matters because it can reduce both **accuracy** and **probability calibration**.
- **Calibration** refers to how well predicted probabilities agree with actual outcome frequencies.

Data Set Shift

- Need: Assumptions about how distribution changes
- Data: X (features); Y (labels); P (Source); Q (Target)
- Covariate Shift Assumption: $p(x) \neq q(x)$, but $p(y | x) = q(y | x)$



Distribution of
source data



Distribution of
target data

- Label Shift Assumption: $p(y) \neq q(y)$, but $p(x | y) = q(x | y)$

Label Shift

$$p(y) \neq q(y), \text{ but } p(x | y) = q(x | y)$$

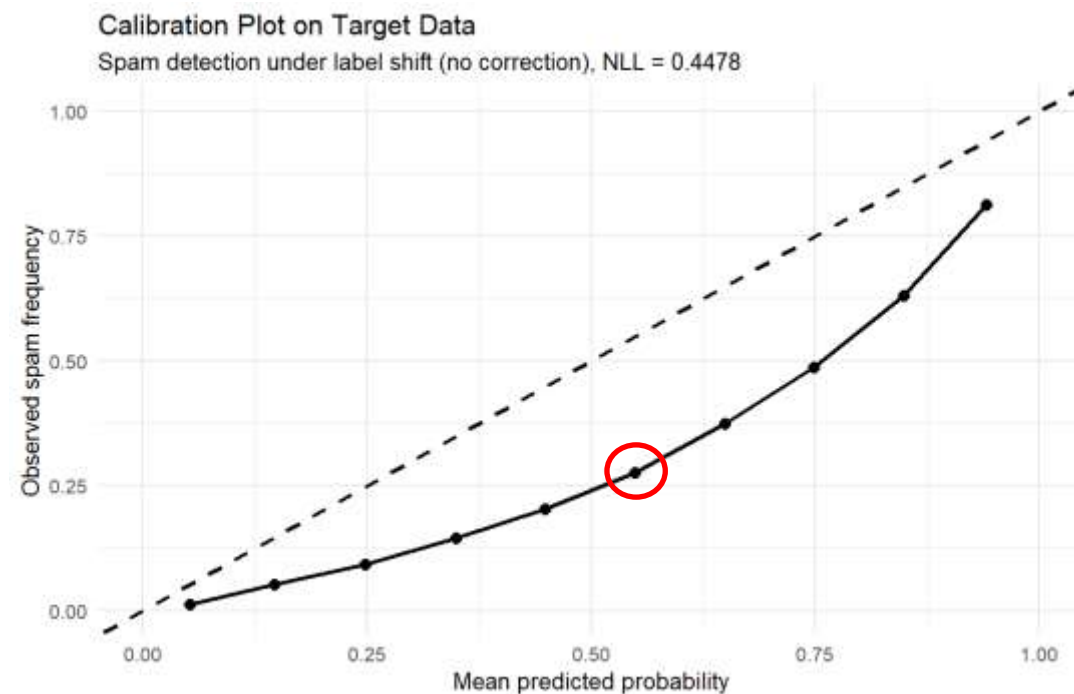
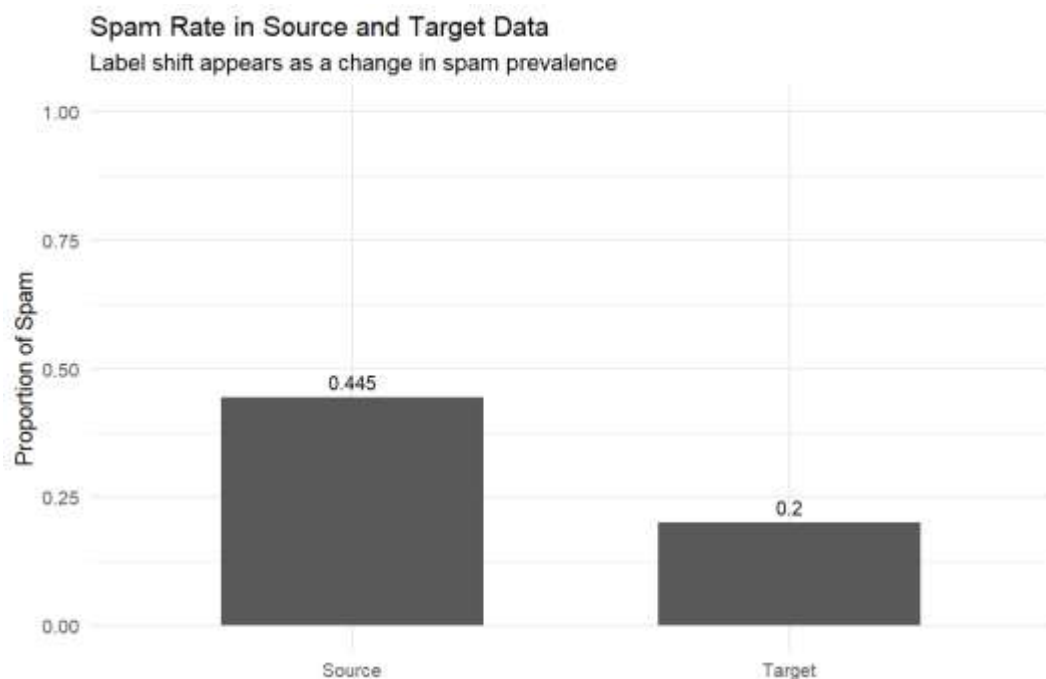
Our Goal: Creating generalizable classification model by selecting stable features that remain reliable under label shift.

Why Label Shift: 1. Current works on data shift mostly focus on covariate shift, but few on label shift.

2. We have a lot of real-world settings where label shift assumption can be applied to.

Eg. Spam detection: patterns of words are similar, but spam rate is different.

Data Simulation:



Estimate Label Shift

However, in practice, we can not observe labels in target data.

Can we estimate $q(y)$? Here are existing literatures:

- Black Box Shift Estimator ¹
 - uses confusion matrix + target prediction frequencies.
- Maximum Likelihood Label Shift ²
 - uses likelihood-based estimation with EM algorithm.

Then we can correct label shift

1: Lipton, Zachary C., Yu-Xiang Wang, and Alex Smola. 2018. "Detecting and Correcting for Label Shift with Black Box Predictors." *arXiv [Cs.LG]*. arXiv. <http://arxiv.org/abs/1802.03916>.

2: Garg, Saurabh, Yifan Wu, Sivaraman Balakrishnan, and Zachary C. Lipton. 2020. "A Unified View of Label Shift Estimation." *arXiv [Cs.LG]*. arXiv. <http://arxiv.org/abs/2003.07554>.

Black Box Shift Estimation (BBSE)

Define: $w_1 = \frac{q(y = 1)}{p(y = 1)}, w_0 = \frac{q(y = 0)}{p(y = 0)}$

In vector form: $\begin{bmatrix} q(\hat{y} = 1) \\ q(\hat{y} = 0) \end{bmatrix} = \begin{bmatrix} p(\hat{y} = 1, y = 1)w_1 & p(\hat{y} = 1, y = 0)w_0 \\ p(\hat{y} = 0, y = 1)w_1 & p(\hat{y} = 0, y = 0)w_0 \end{bmatrix}$

$$\begin{bmatrix} \mu \end{bmatrix} = \begin{bmatrix} C \end{bmatrix} \begin{bmatrix} w \end{bmatrix}$$

$$\begin{bmatrix} C^{-1} \end{bmatrix}^{-1} \begin{bmatrix} \mu \end{bmatrix} = \begin{bmatrix} w \end{bmatrix}$$

BBSE Example

$$w_1 = \frac{q(y=1)}{p(y=1)}, w_0 = \frac{q(y=0)}{p(y=0)}$$

Suppose there are 2 classes: $y=0$ (not spam) and $y=1$ (spam)

On Source: 50% spam and 50% not spam; On Target: 60% spam and 40% not spam

True Label	True $y = 1$	True $y = 0$	Total
Predicted $y = 1$	40	10	50
Predicted $y = 0$	10	40	50
Total	50	50	100

So $\begin{bmatrix} w_1 \\ w_0 \end{bmatrix} = \begin{bmatrix} 1.2 \\ 0.8 \end{bmatrix}$. Imagine we predict on target data set and get $\begin{bmatrix} q(\hat{y}=1) \\ q(\hat{y}=0) \end{bmatrix} = \begin{bmatrix} 0.55 \\ 0.45 \end{bmatrix}$,

we have $\begin{bmatrix} 0.4 & 0.1 \\ 0.1 & 0.4 \end{bmatrix}^{-1} \begin{bmatrix} 0.55 \\ 0.45 \end{bmatrix} = \begin{bmatrix} \hat{w}_1 \\ \hat{w}_0 \end{bmatrix}$.

$$\frac{\begin{bmatrix} 0.4 & -0.1 \\ -0.1 & 0.4 \end{bmatrix}}{0.4 \times 0.4 - 0.1 \times 0.1} \begin{bmatrix} 0.55 \\ 0.45 \end{bmatrix} = \frac{\begin{bmatrix} 0.4 & -0.1 \\ -0.1 & 0.4 \end{bmatrix}}{0.15} \begin{bmatrix} 0.55 \\ 0.45 \end{bmatrix} = \frac{1}{0.15} \begin{bmatrix} 0.175 \\ 0.125 \end{bmatrix} = \begin{bmatrix} 1.16 \\ 0.83 \end{bmatrix} \approx \begin{bmatrix} 1.2 \\ 0.8 \end{bmatrix}$$

What if the matrix **NOT** invertible?

Maximum Likelihood Label Shift (MLLS)

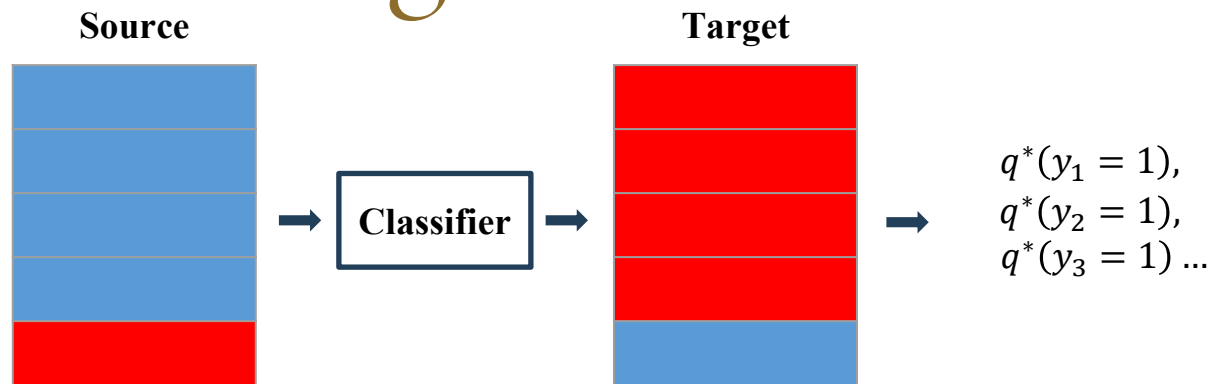
- MLLS approaches this problem from a **likelihood perspective**: it looks for weights that make the reweighted source model fit the unlabeled target data well.
- This leads to an optimization problem, rather than a linear system like BBSE.
- The target labels are **not observed**, so the optimization is naturally an **incomplete-data** problem.
- Therefore, a natural tool for this setting is the **EM algorithm**¹, which is designed for maximum likelihood estimation with missing or latent variables.

1. Garg, Saurabh, Yifan Wu, Sivaraman Balakrishnan, and Zachary C. Lipton. 2020. “A Unified View of Label Shift Estimation.” *arXiv [Cs.LG]*. arXiv. <http://arxiv.org/abs/2003.07554>.

Saerens, Marco, Patrice Latinne, and Christine Decaestecker. 2002. “Adjusting the Outputs of a Classifier to New a Priori Probabilities: A Simple Procedure.” *Neural Computation* 14 (1): 21–41.

Dempster, A. P., N. M. Laird, and D. B. Rubin. 1977. “Maximum Likelihood from Incomplete Data via the EM Algorithm.” *Journal of the Royal Statistical Society* 39 (1): 1–38.

EM Algorithm



Update Rule: $q^{(t+1)}(y = 1) = \frac{1}{n} \sum_{i=1}^n r_{i1}^{(t)}$,
 where $n = \text{sample size}$

$$r_{i1}^{(t)} = \frac{q^*(y = 1) \times \frac{q^{(t)}(y = 1)}{p(y = 1)}}{\sum_{\ell=0}^k q^*(y = \ell) \frac{q^{(t)}(y = \ell)}{p(y = \ell)}},$$

where $k = \# \text{ of labels}$

Initialize: $q^0(y = 1) = p(y = 1)$

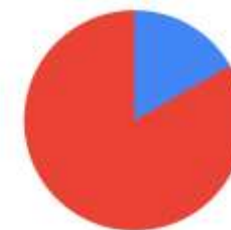


Update: $q^{(t+1)}(y = 1) = \frac{1}{n} \sum_{i=1}^n r_{i1}^{(t)}$

$\vdots$



Repeat Until Convergence:

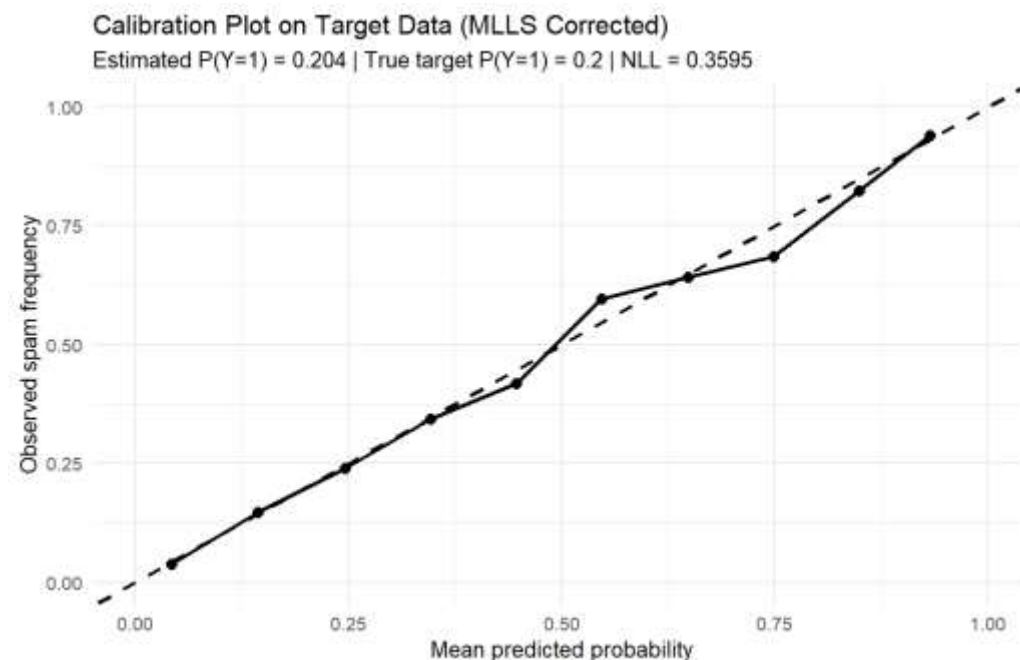
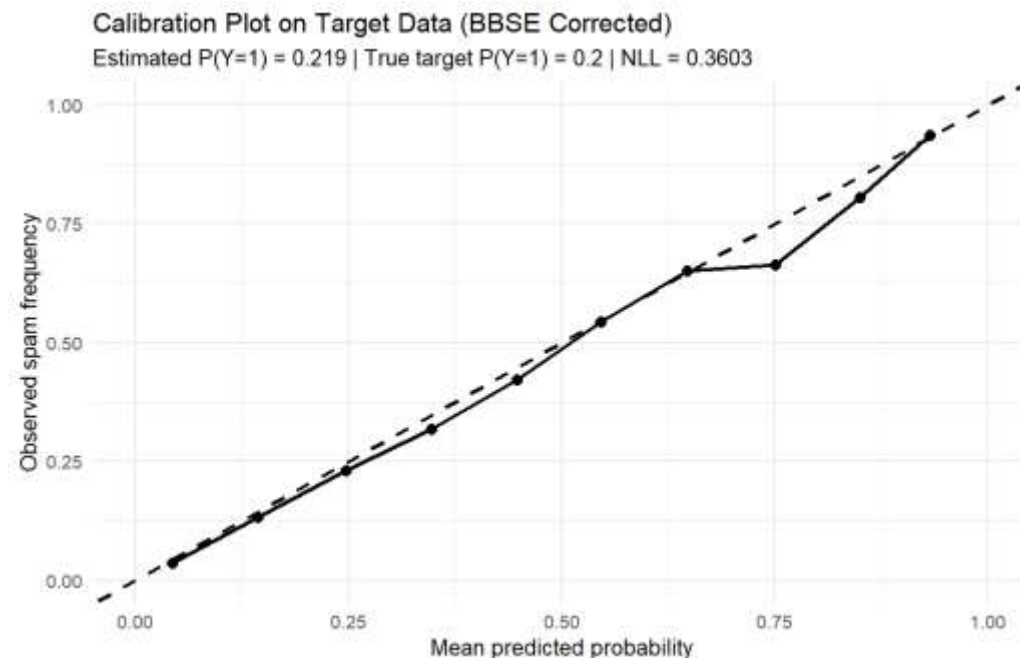
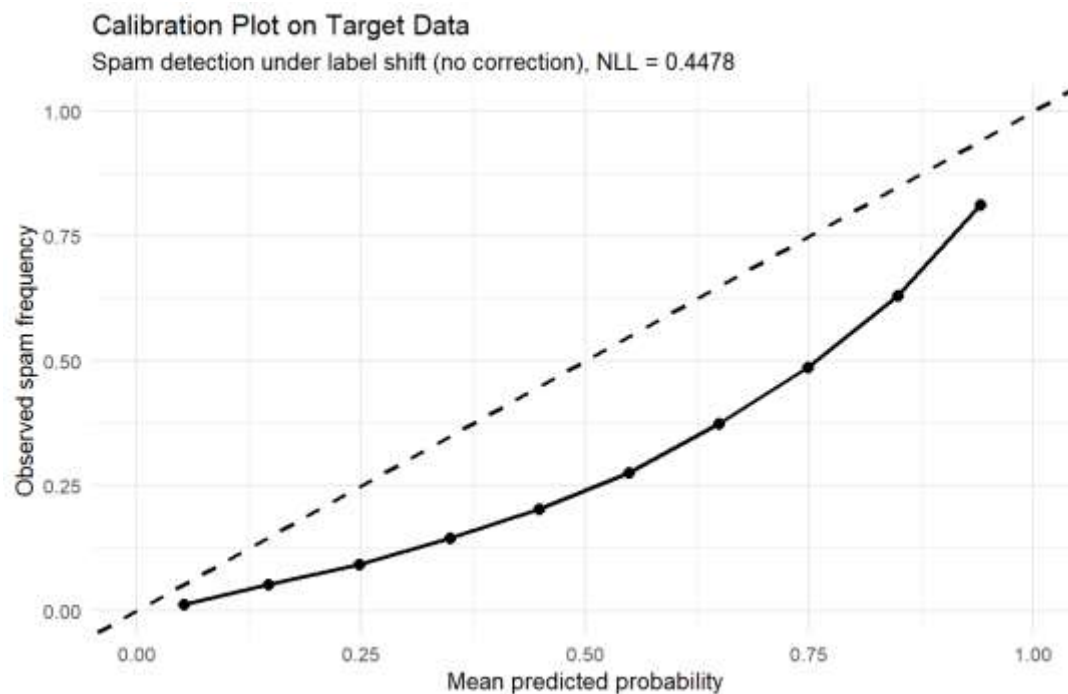


Calibration

Let $p(1) := p(y = 1 | x)$

$$\hat{q}_{\text{BBSE}}(y = 1 | x) = \frac{\hat{w}_1^{\text{BBSE}} p(1)}{\hat{w}_1^{\text{BBSE}} p(1) + \hat{w}_0^{\text{BBSE}} [1 - p(1)]}$$

$$\hat{q}_{\text{MLLS}}(y = 1 | x) = \frac{\hat{w}_1^{\text{MLLS}} p(1)}{\hat{w}_1^{\text{MLLS}} p(1) + \hat{w}_0^{\text{MLLS}} [1 - p(1)]}$$



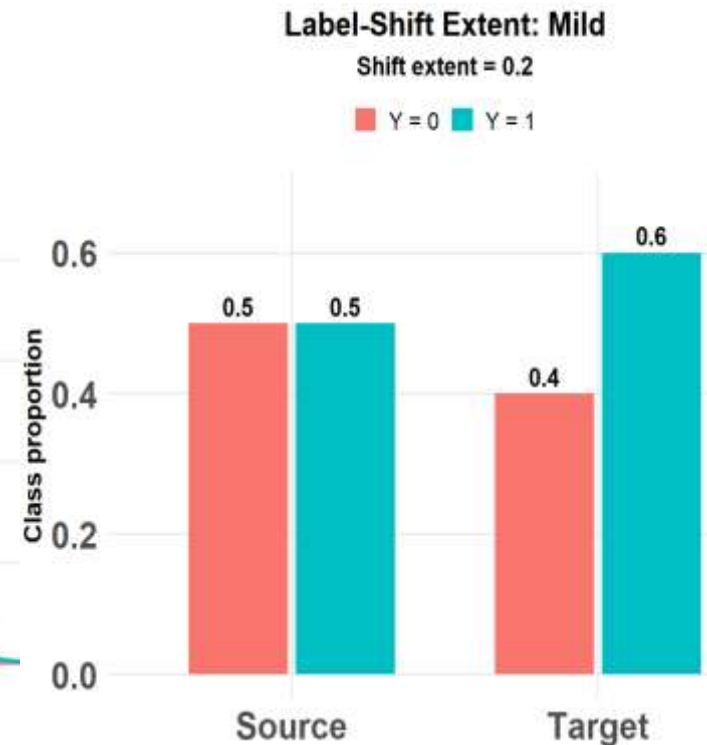
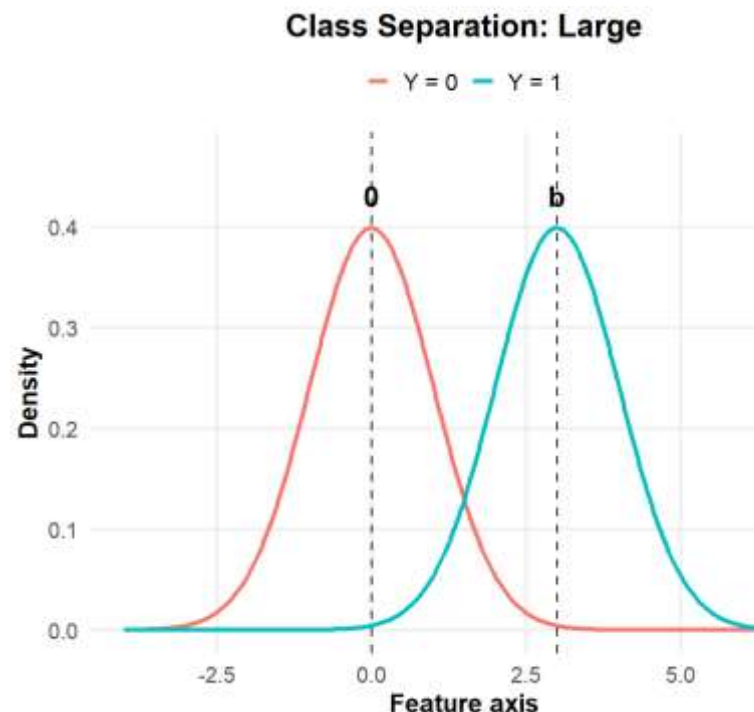
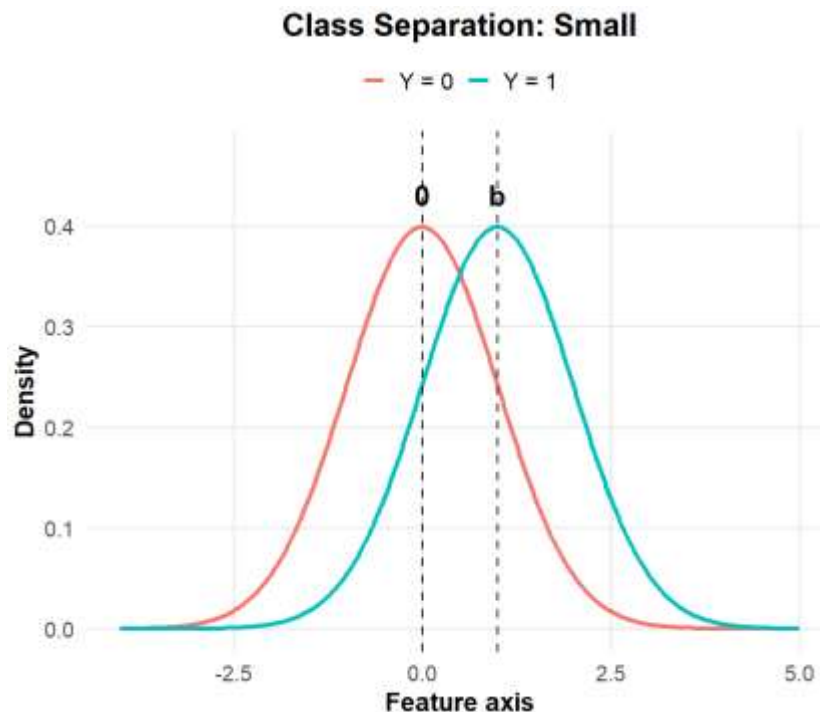
Comparison between BBSE and MLL

We compare BBSE and MLLS in binary cases for different:

(1) class separation, (2) label-shift extent, (3) sample size

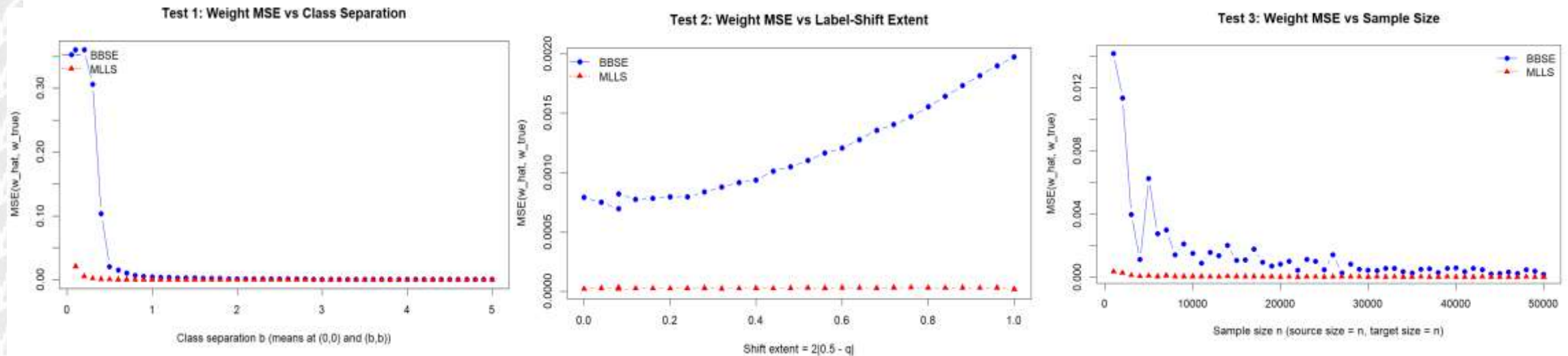
Class Separation: Distance between conditional distribution $p(x|y)$

Label Shift Extent: Difference between $p(y)$ and $q(y)$.



Comparison between BBSE and MLLS

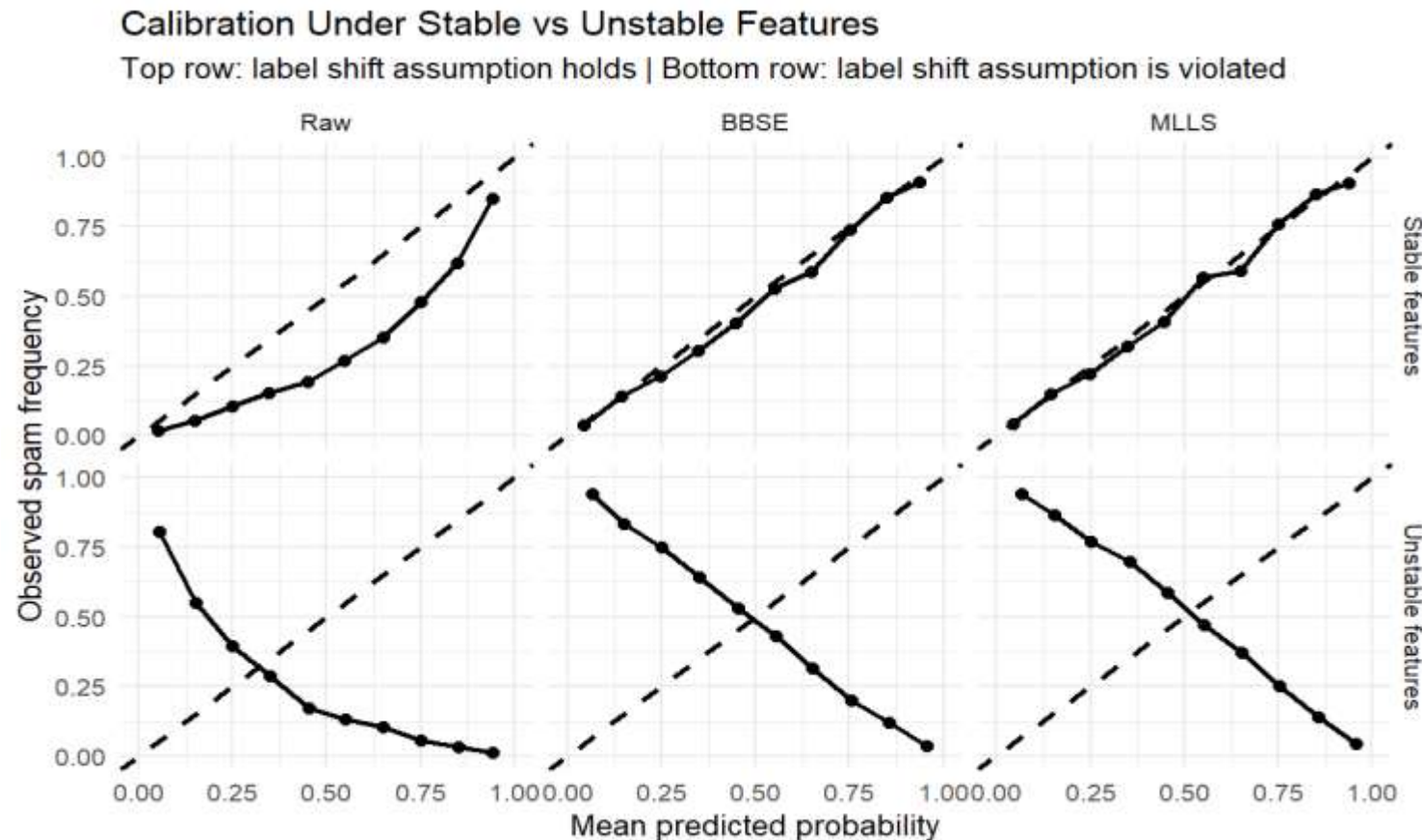
Performance metric: **weight estimation error**, measured by **MSE** between estimated weight and true weight. Lower is better.



Stable & Unstable Variables

So far we only have features satisfying label shift, what if not?

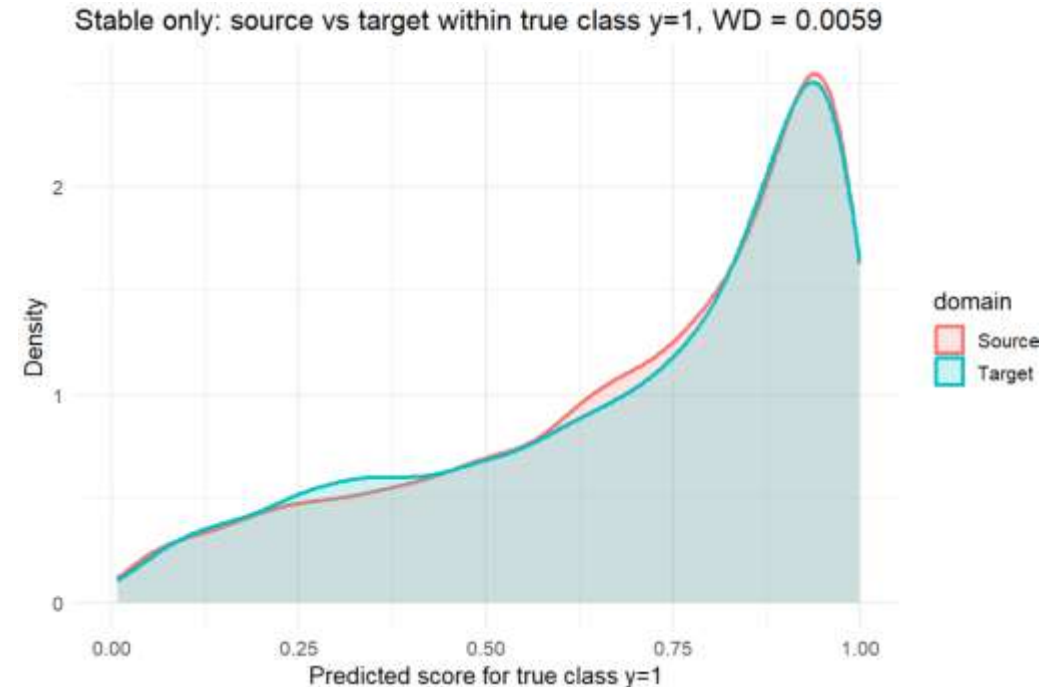
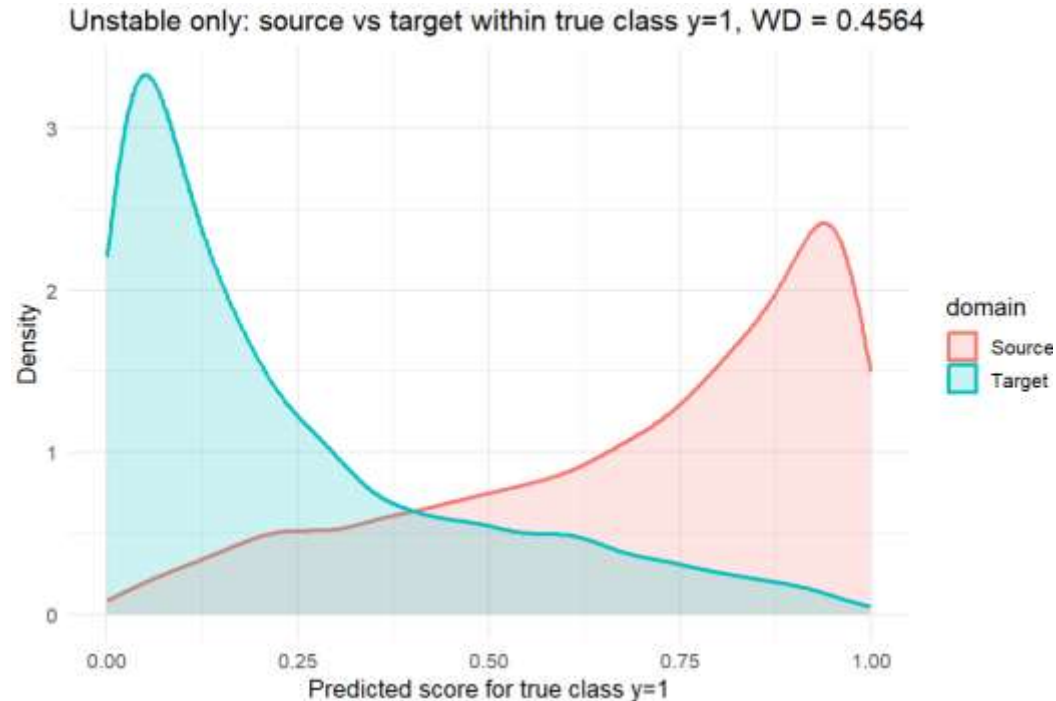
- Stable Features: Features satisfy label shift assumption
- Unstable Features: Features do not satisfy label shift assumption



Variable Selection

So, how can we select stable features?

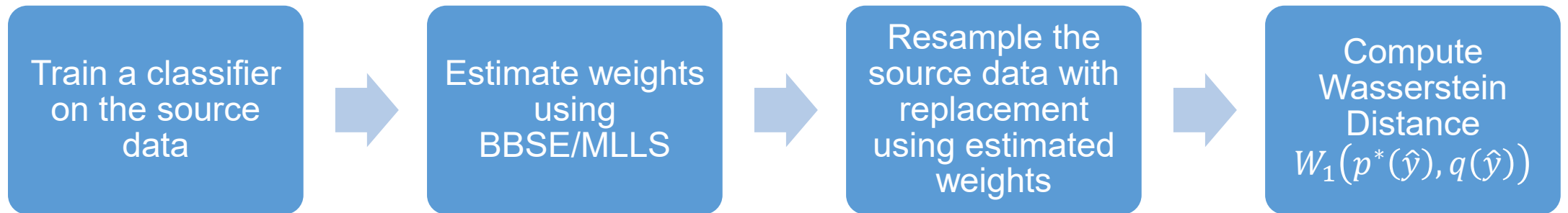
- Wasserstein Distance $W_1(P, Q) = \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)| du$
- It measures how close two distributions are from each other.



Resampling Procedure

- However, we do not know target labels.
- Instead, we can measure $W_1(p^*(\hat{y}), q(\hat{y}))$, where $p^*(\hat{y})$ is the resampled source distribution based on *estimated weights*.
- A **large Wasserstein distance** suggests the feature set is **Unstable**.

Here is how we resample:



Criterion

Let us define our criterion to select variable be:

$$\text{Criterion} = \text{NLL} + \lambda \cdot W_1(p^*(\hat{y}), q(\hat{y})),$$

where $p^*(\hat{y})$ is the resampled distribution of predicted scores in the source data after reweighting by BBSE/MLLS estimate, and $q(\hat{y})$ is the distribution of predicted scores in the target data.

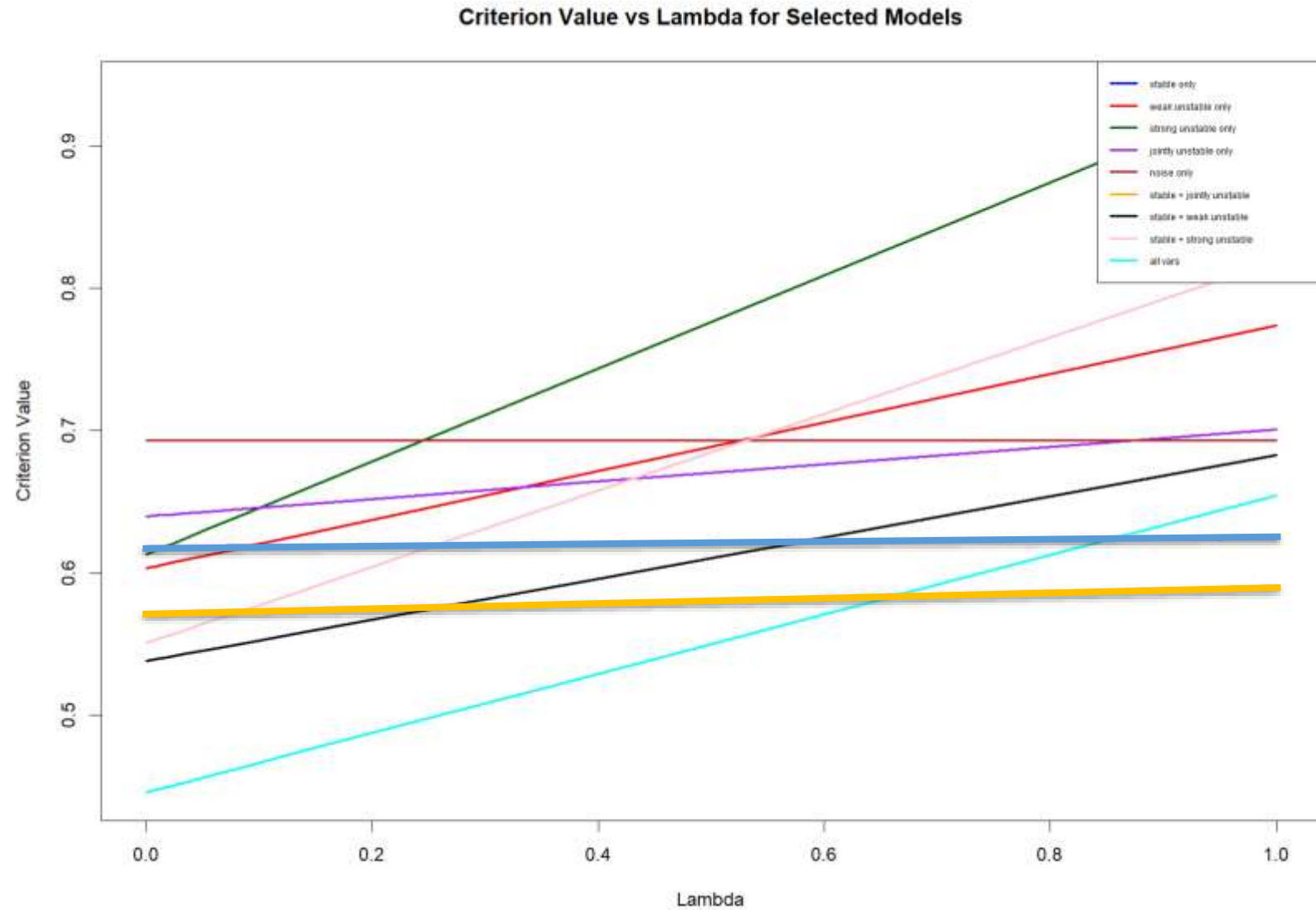
So, as λ getting larger, stable features will be more advantaged in the selection process.

Simulation

In order to see which features will be selected, we simulated the following experiments:

- Source data with $n = 6000$, Target data with $n = 3000$
- $p(y = 1) = 0.5$ for source; $q(y = 1) = 0.7$ for target
- Features are
 - Stable
 - Weakly unstable
 - Strongly unstable
 - Marginally stable, jointly unstable
 - Noise

Variable Selection Result



Conclusion & Discussion

- We can use BBSE/MLLS to improve the model as long as our features are stable.
- MLLS is slightly better and more stable than BBSE.
- Wasserstein Distance is a useful tool to detect stable features.
- Unstable features are not 100% bad in terms of prediction, but enough stable features are needed for a well calibrated model.

Thank you

Black Box Shift Estimation (BBSE)

We want: $w_1 = \frac{q(y = 1)}{p(y = 1)}$

$$\begin{aligned} q(\hat{y} = 1) &= q(\hat{y} = 1, y = 1) + q(\hat{y} = 1, y = 0) \\ &= q(\hat{y} = 1 \mid y = 1)q(y = 1) + q(\hat{y} = 1 \mid y = 0)q(y = 0) \end{aligned}$$

By Label Shift Assumption, we have $q(\hat{y} = 1 \mid y = 1) = p(\hat{y} = 1 \mid y = 1)$, and

$$q(\hat{y} = 1 \mid y = 0) = p(\hat{y} = 1 \mid y = 0)$$

$$\begin{aligned} \text{SO, } q(\hat{y} = 1) &= p(\hat{y} = 1 \mid y = 1)q(y = 1) + p(\hat{y} = 1 \mid y = 0)q(y = 0) \\ &= \frac{p(\hat{y} = 1, y = 1)}{p(y = 1)}q(y = 1) + \frac{p(\hat{y} = 1, y = 0)}{p(y = 0)}q(y = 0) \\ &= p(\hat{y} = 1, y = 1)w_1 + p(\hat{y} = 1, y = 0)w_0 \end{aligned}$$

Similarly, $q(\hat{y} = 0) = p(\hat{y} = 0, y = 1)w_1 + p(\hat{y} = 0, y = 0)w_0$

EM Algorithm & MLLS

Steps to use EM Algorithm

- Train a probabilistic classifier on the source data and predict on the target data.
- Initialize the target class proportions using the source class proportions.
- compute soft labels for target observations.
- update the target class proportions.
- Repeat until convergence, then convert the result into label-shift weights.

Label Shift Correction

After we estimate the weight, how can we correct for label shift?

- With Bayes' Rule, we have $q(y = j | x) = \frac{q(y=j) q(x|y = j)}{q(x)}$
- Under Label Shift: $q(y = j | x) = \frac{q(y=j) p(x|y = j)}{\sum_{\ell=1}^k q(y=\ell) p(x|y = \ell)}$
- Since $p(y = j | x) = \frac{p(y=j) p(x|y = j)}{p(x)}$, we can rewrite the target posterior as $\frac{w_j p(y = j|x)}{\sum_{\ell=1}^k w_{\ell} p(y = \ell|x)}$
- Thus, once we have the estimated weights, we can plug it back.
- Let $p(1) := p(y = 1 | x)$. In binary cases, we have:

$$\hat{q}_{\text{BBSE}}(y = 1 | x) = \frac{\hat{w}_1^{\text{BBSE}} p(1)}{\hat{w}_1^{\text{BBSE}} p(1) + \hat{w}_0^{\text{BBSE}} [1 - p(1)]}$$

$$\hat{q}_{\text{MLLS}}(y = 1 | x) = \frac{\hat{w}_1^{\text{MLLS}} p(1)}{\hat{w}_1^{\text{MLLS}} p(1) + \hat{w}_0^{\text{MLLS}} [1 - p(1)]}$$